PHYSICAL CHEMISTRY CUMULATIVE EXAM

November 16, 1991

Classical physical chemistry, and science in general, abounds with concepts, rules, equations, and laws which are attributed to their authors or discoverers. Describe/define as many of the following classical notions as time permits. An equation *per se* is not sufficient; description of variables and their meaning is also required; plots or diagrams are good descriptive aids.
[Passing score: any 12 correct]

Trouton's rule

Law of Dulong and Petit

Clapeyron equation

Clausius-Clapeyron equation

Joule-Thomson effect

Kirchoff's law (enthalpy)

Lindemann-Hinshelwood mechanism

Michaelis-Menten mechanism

Law of Charles/Gay-Lussac

Maxwell-Boltzmann distribution

Helmholtz function

Henderson-Hasselbalch equation

Lineweaver-Burk plot

London force

Henry's law

Van't Hoff relation

van der Waals force

Arrhenius Law

Carnot cycle

Debye-Huckel theory

Faraday constant

Eyring equation

RRKM theory

Hess's law

Gibbs function

Born-Haber cycle

Nernst equation

Boyle's law

Fick's law

Raoult's law